

Abstract No.

Characterisation of calcium active tumor microtubule networks in diffuse midline glioma in multiple models

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BACKGROUND: Diffuse midline glioma (DMG) is an aggressive and resistant paediatric-type high-grade glioma. Its growth is influenced in part by neuronal activity and related signalling pathways. In glioblastoma, tumor microtubes (TMs), long cellular extensions, interconnect tumour cells to dense networks. Within these networks, pacemaker-like cells trigger hierarchical rhythmic calcium signalling. This connectivity and communication drive tumour growth and resistance to standard-of-care treatments. However, it is not yet fully understood whether DMGs share these same connective and communicative features.

METHODS: Using several patient-derived DMG cell lines, we examined TM-based cell networks across a range of model systems — 2D monolayers, spheroids, co-cultures with iPSC-derived human cortical organoids, and mouse xenografts — and confirmed key findings in human tumour tissue.

Immunostaining, live calcium imaging, and two-photon in vivo microscopy were used to characterise the structure and activity of these networks.

RESULTS: Nestin-positive TMs forming tumor cell networks were consistently identified in DMG across cell cultures, xenograft models, and patient specimens. Network morphology varied among different patient-derived lines, differing in TM length, branching patterns, and proximity to blood vessels. Distinct morphological types resembled radial glia, neurons, perivascular cells, and OPCs across experimental settings. Observing DMG invasion in vivo over multiple weeks using two-photon microscopy showed single-cell infiltration patterns similar to neuronal migration and invasion patterns seen in glioblastoma. Calcium imaging in vitro and in vivo revealed calcium activity within DMG networks including waves and repeated rhythmic bursts, supporting functional connections between tumor cells.

CONCLUSIONS: This work characterises TM-driven network architecture and calcium signalling activity in DMG across diverse experimental models, suggesting these networks could serve as future therapeutic targets. The ability to observe these processes directly in vivo offers a valuable tools of studying and manipulating DMG biology in the living brain.



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